Project Title: Predict New York City Airbnb Prices Using Linear Regression

The goal of this project was to predict the price of Airbnb listings in New York City using a Linear Regression model. First, the dataset was explored and cleaned. Missing values were handled, unnecessary columns were removed, outliers were treated, and categorical columns were converted into numerical values using One-Hot Encoding. After that, the data was divided into training and testing sets. The Linear Regression model was then trained and tested.

The model achieved an **R^2 Score of 0.4708277350002139** and a **Mean Absolute Error (MAE) of 38.5408749155396** .

The **R^2 Score** shows how well the model predicts Airbnb prices. For example, if the R^2 Score is **0.47**, it means the model can explain **47% of the changes in Airbnb prices**. A higher R^2 Score means the model performs better.

The **MAE (Mean Absolute Error)** shows the average difference between the actual price and the predicted price. For example, if the MAE is **38.54**, it means the model's prediction is, on average, **$38.54 away from the actual price**. A smaller MAE means the model is more accurate.

The model works better for **cheap and medium-priced Airbnb listings** because there are more of these listings in the dataset. It is less accurate for **expensive listings** because they have fewer examples and their prices vary a lot.

Conclusion:

The Linear Regression model gives a good estimate of Airbnb prices and can help understand price trends. However, more advanced machine learning models may provide better accuracy, especially for expensive properties.